

Shuyang Zhu (朱舒阳)

Industrial CT · Geometry Processing · CUDA-Acceleration
derulozsy@gmail.com | GitHub | Homepage

Profile

Graduate student at the University of Tokyo working at the intersection of industrial CT reconstruction, 3D geometry processing, and CUDA acceleration. My research interests center on Surface Modeling and high-performance CT Reconstruction.

Publications

An automatic framework for quadrilateral surface reconstruction with partitions from 3D point clouds

Shuyang Zhu, Youlong Li, Jizhao Liu, Bin Li*.

The Visual Computer.

Proposes a four-step framework inspired by polycube parameterization to reconstruct 3D point clouds into quadrilateral meshes with smooth partitions while preserving geometric features.

CTGS: Hybrid Gaussian Splatting for Compact Industrial CT Representation

Shuyang Zhu*, Yukie Nagai.

Accepted by ICPE 2026.

Represents industrial CT volumes with Gaussian Splatting using CT-specific slice supervision, geometric regularization, and CUDA acceleration for volume-based training and analysis.

Selected Projects

Adaptive CT Volume

Ongoing

Adaptive projection-domain CT reconstruction

- Modeled CT volumes as anisotropic octrees for compression and optimized the representation directly from sparse projection data.
- Maintains clear edge features and accurate surface extraction under sparse-projection training; compressed model size is around 8 MB.

CT Simulation Software

Ongoing

High-accuracy and customizable CT simulation

- Built software to simulate CT scans from 3D models with physical accuracy.
- Included lattice generation, defect injection, physics-based compression simulation using FEM, and GPU dynamic batching for large-volume processing.

Porous CT Processor

Ongoing

Pore network modeling + 4D CT analysis

- Designed an analysis and visualization suite with GUI for porous CT datasets.
- Integrated rendering, pore network modeling, 4D tracking of pore evolution, and GPU acceleration using CuPy.

STL Processor

DirectX 12 mesh processing and visualization

- Developed a DirectX 12 application for STL mesh repair and visualization.
- Implemented polycube parameterization, hole filling, normal correction, denoising, and smoothing.

Work Experience

Tencent

Aug 2026 - Present

Game Research & Development Intern, Engine Research

Polyphony Digital

Jul 2026

Engineer Intern

- Developed a 3D Gaussian Splatting segmentation and extraction tool.

Synesthesia

May 2026 - Present

Technical Development Intern

- Measured wall-surface inclination from complex point clouds with millimeter-level precision.

Fixstars

Apr 2026 - May 2026

Year-round Intern

- Optimized VRAM management for video generation models.

Education

The University of Tokyo, Tokyo, Japan <i>M.E. candidate, Nuclear Engineering and Management</i>	Oct 2025 - Present
Hokkaido University, Sapporo, Japan <i>Special Auditor, Department of Electronics and Information Engineering</i>	Oct 2024 - Mar 2025
Lanzhou University, Lanzhou, China <i>B.E. in Digital Media Technology</i>	Sep 2021 - Jul 2025

Awards

- 2025:** The University of Tokyo Fellowship
- 2023:** 9th Internet+ Innovation and Entrepreneurship Competition, Gansu Province Gold Award
- 2022:** 13th BlueBridge Cup, Gansu Province Third Prize
- 2022-2024:** Lanzhou University Scholarship